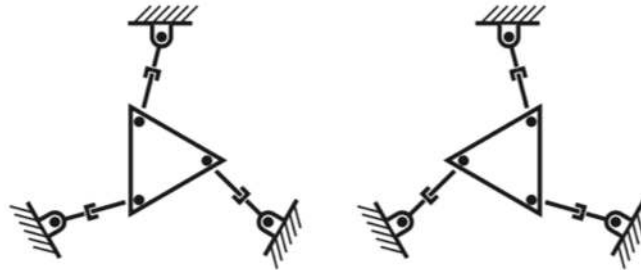


Chapter 2	Configuration Space
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

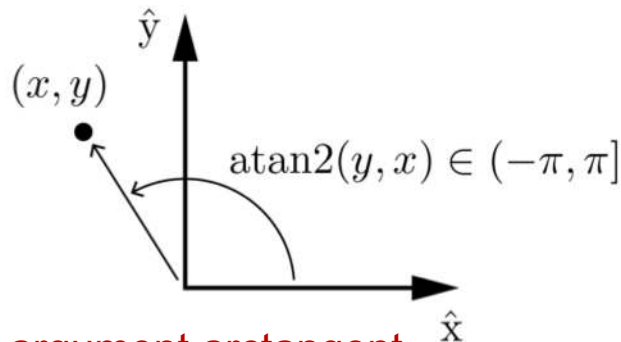
Important concepts, symbols, and equations

- **inverse kinematics (IK)**: given $T_{sd} \in SE(3)$, find θ such that $T(\theta) = T_{sd}$.
- unlike FK, IK for serial chains could have zero, one, or multiple solutions

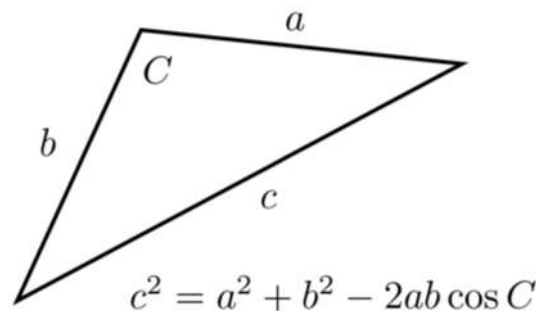
situation is reversed
for closed chains:



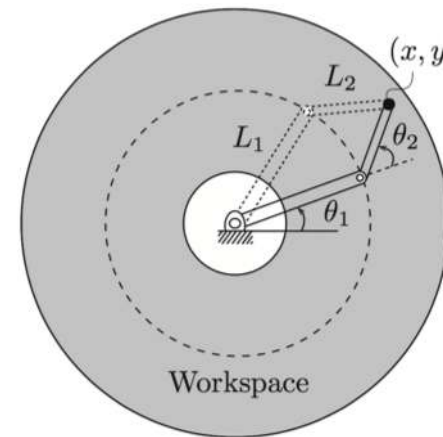
- closed-form (analytical) IK



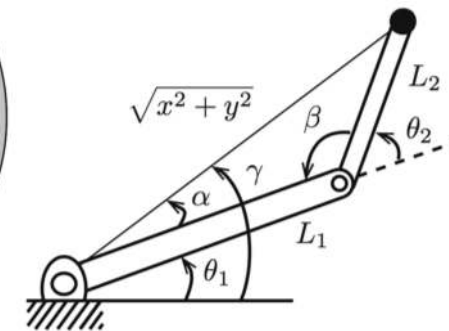
two-argument arctangent



law of cosines



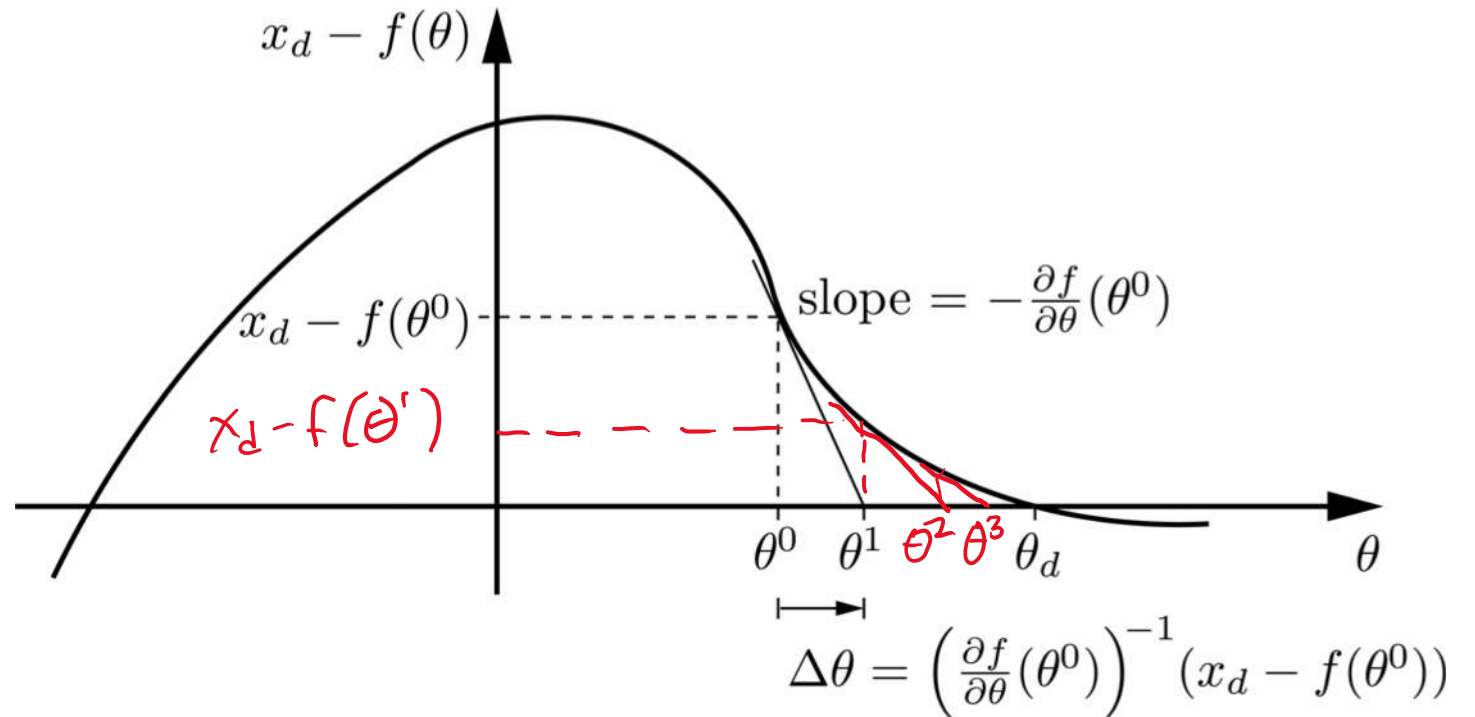
IK for a 2R robot
 γ from atan2
 α, β from law of cosines



Important concepts, symbols, and equations (cont.)

- **numerical IK**: iteratively refine an initial guess θ^0 to find θ^k such that $T(\theta^k) \approx T_{sd}$.
- **Newton-Raphson root finding**

x_d : desired function value
 $f(\theta)$: actual function value at θ



Important concepts, symbols, and equations (cont.)

- vector Taylor expansion: $x_d = f(\theta_d) = f(\theta^0) + \underbrace{\frac{\partial f}{\partial \theta} \Big|_{\theta^0}}_{J(\theta^0)} \underbrace{(\theta_d - \theta^0)}_{\Delta \theta} + \text{h.o.t.}$

linear approximation: $J(\theta^0) \Delta \theta = x_d - f(\theta^0)$ $Ax = b$ (1)

linear correction to the guess θ^0 : $\Delta \theta = J^{-1}(\theta^0) (x_d - f(\theta^0))$

- if J is not invertible, use the **pseudoinverse**: $\Delta \theta^* = J^\dagger(\theta^0) (x_d - f(\theta^0))$

If there exists a $\Delta \theta$ exactly satisfying (1), then $\Delta \theta^*$ has the smallest 2-norm among all solutions.

If there is no $\Delta \theta$ exactly satisfying (1), then $\Delta \theta^*$ comes closest in the 2-norm sense.

Important concepts, symbols, and equations (cont.)

Special cases of pseudoinverse for $J \in \mathbb{R}^{m \times n}$ (m e-e velocity directions, n joints):

- If J is full rank and square: $J^\dagger = J^{-1}$
 $m=6$ for twist
- If J is full rank and tall ($m > n$): $J^\dagger = (J^T J)^{-1} J^T \in \mathbb{R}^{n \times m}$ (the “left inverse”)
 $J^T J \in \mathbb{R}^{n \times n}$, invertible $J^+ J = I$
- If J is full rank and wide ($n > m$): $J^\dagger = J^T (J J^T)^{-1} \in \mathbb{R}^{n \times m}$ (the “right inverse”)
 $J J^T \in \mathbb{R}^{m \times m}$, invertible $J J^+ = I$

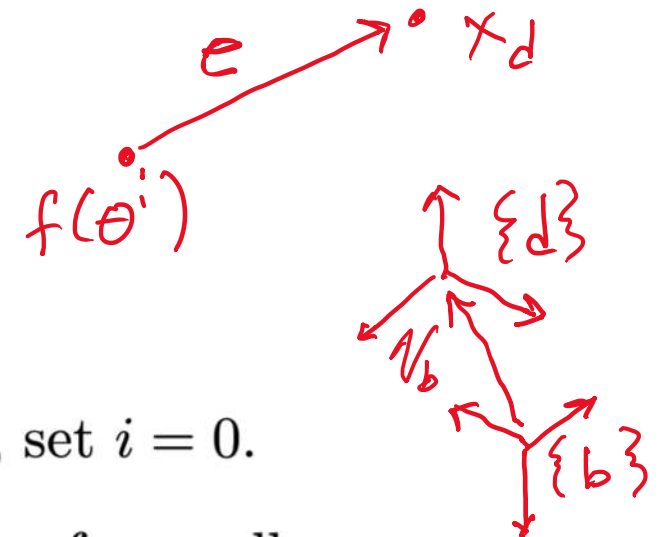
Important concepts, symbols, and equations (cont.)

- Numerical inverse kinematics, coordinate version:

(a) **Initialization:** Given $x_d \in \mathbb{R}^m$ and an initial guess $\theta^0 \in \mathbb{R}^n$, set $i = 0$.

(b) Set $e = x_d - f(\theta^i)$. While $\|e\| > \epsilon$ for some small ϵ :

- Set $\theta^{i+1} = \theta^i + J^\dagger(\theta^i)e$.
- Increment i .



- Numerical inverse kinematics, geometric version:

(a) **Initialization:** Given T_{sd} and an initial guess $\theta^0 \in \mathbb{R}^n$, set $i = 0$.

(b) Set $[\mathcal{V}_b] = \log(T_{sb}^{-1}(\theta^i)T_{sd})$. While $\|\omega_b\| > \epsilon_\omega$ or $\|v_b\| > \epsilon_v$ for small $\epsilon_\omega, \epsilon_v$:

- Set $\theta^{i+1} = \theta^i + J_b^\dagger(\theta^i)\mathcal{V}_b$.
- Increment i .

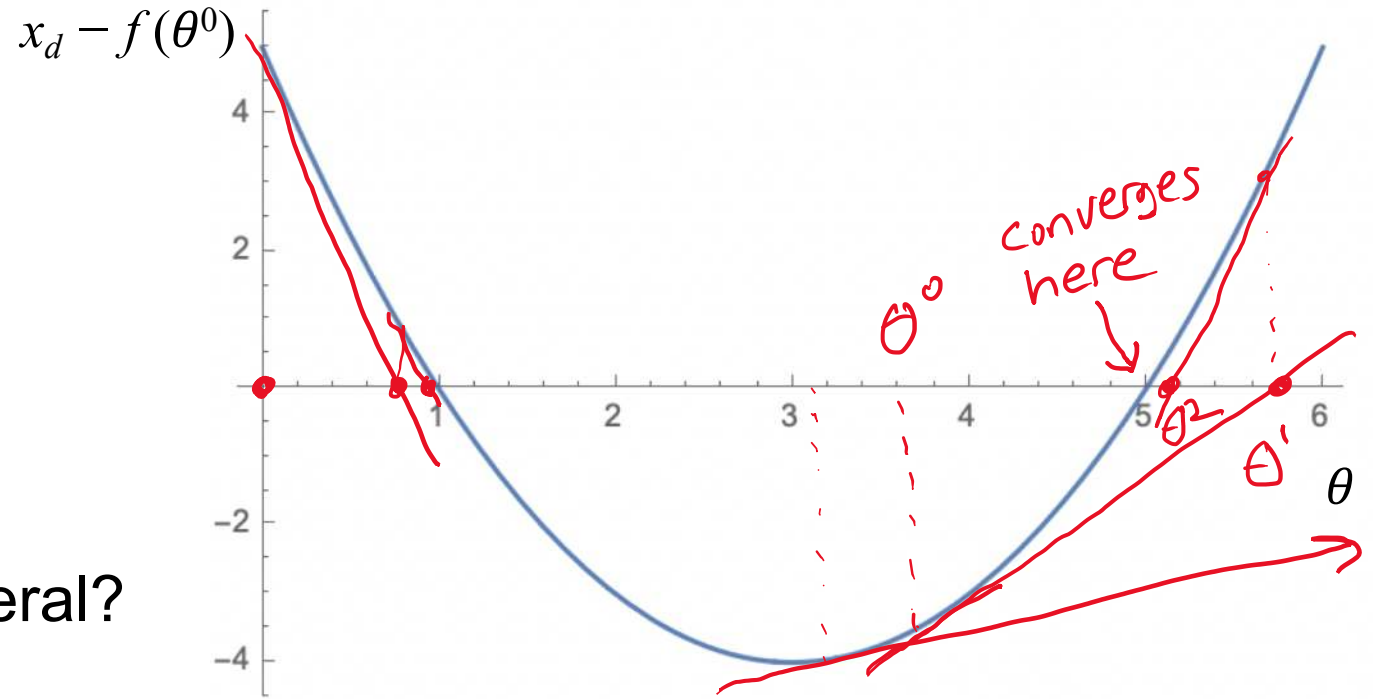
$$[\mathcal{V}_b] = \log(T_{b \cancel{s}}^{-1} T_{\cancel{s}d})$$

Illustrate Newton-Raphson root finding
for initial guesses $\theta^0 = 3.6$ and $\theta^0 = 0$.

What if $\theta^0 = 3.1$ and this were a joint
angle?

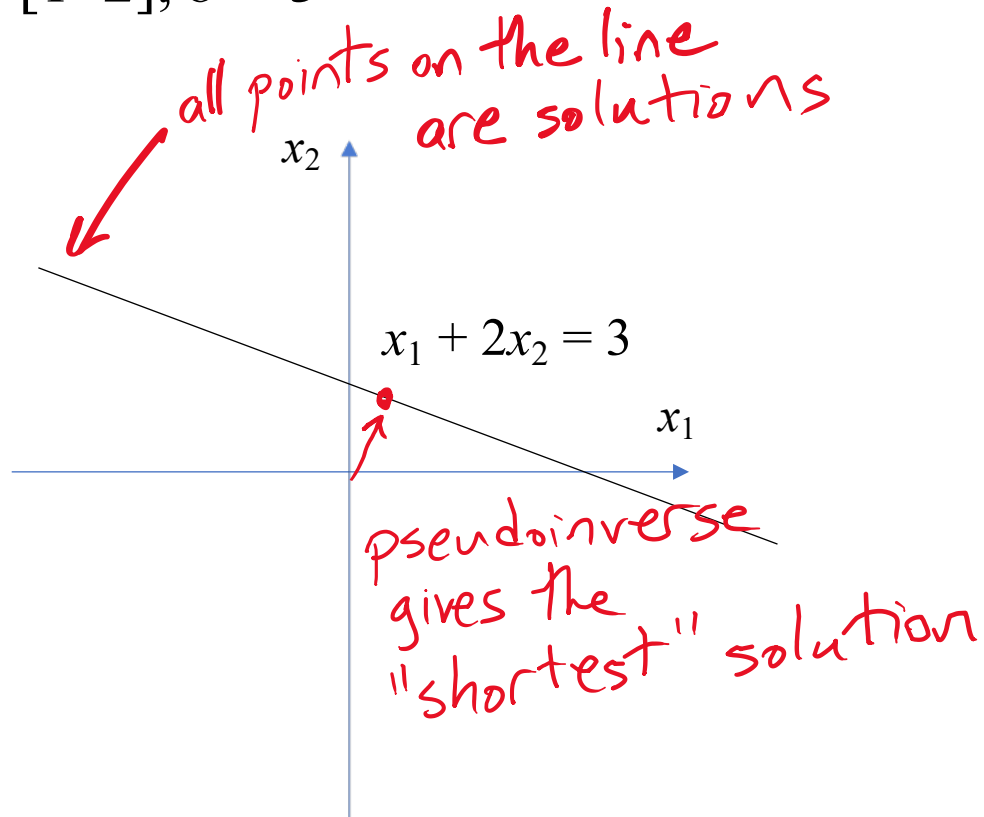
For a robot controller, what's a good
choice for the initial guess θ^0 in general?

*the solution to the IK at
the previous time step*

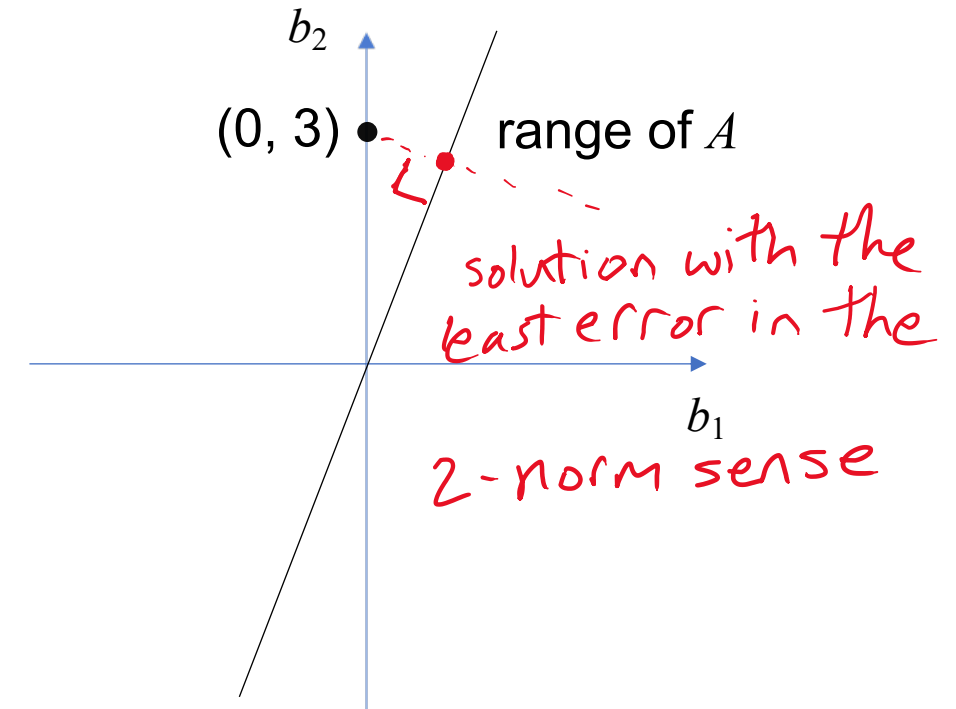


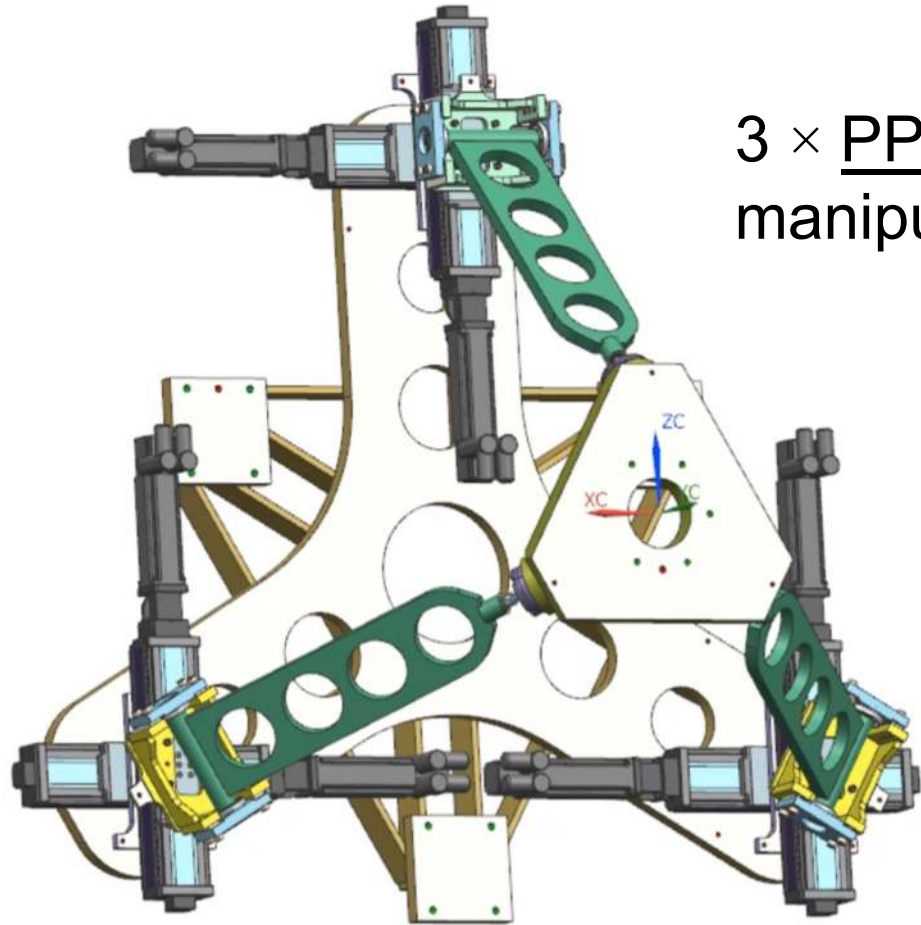
Graphically find a “good” solution to $Ax = b$, e.g., $x = A^\dagger b$.

1) $A = [1 \ 2]$, $b = 3$



2) $A = [1 \ 2]^T$, $b = [0 \ 3]^T$





3 × PPRS parallel
manipulator

6 dof

IK? FK?

easy use Newton-Raphson

KUKA Systems North America LLC
(patent pending)